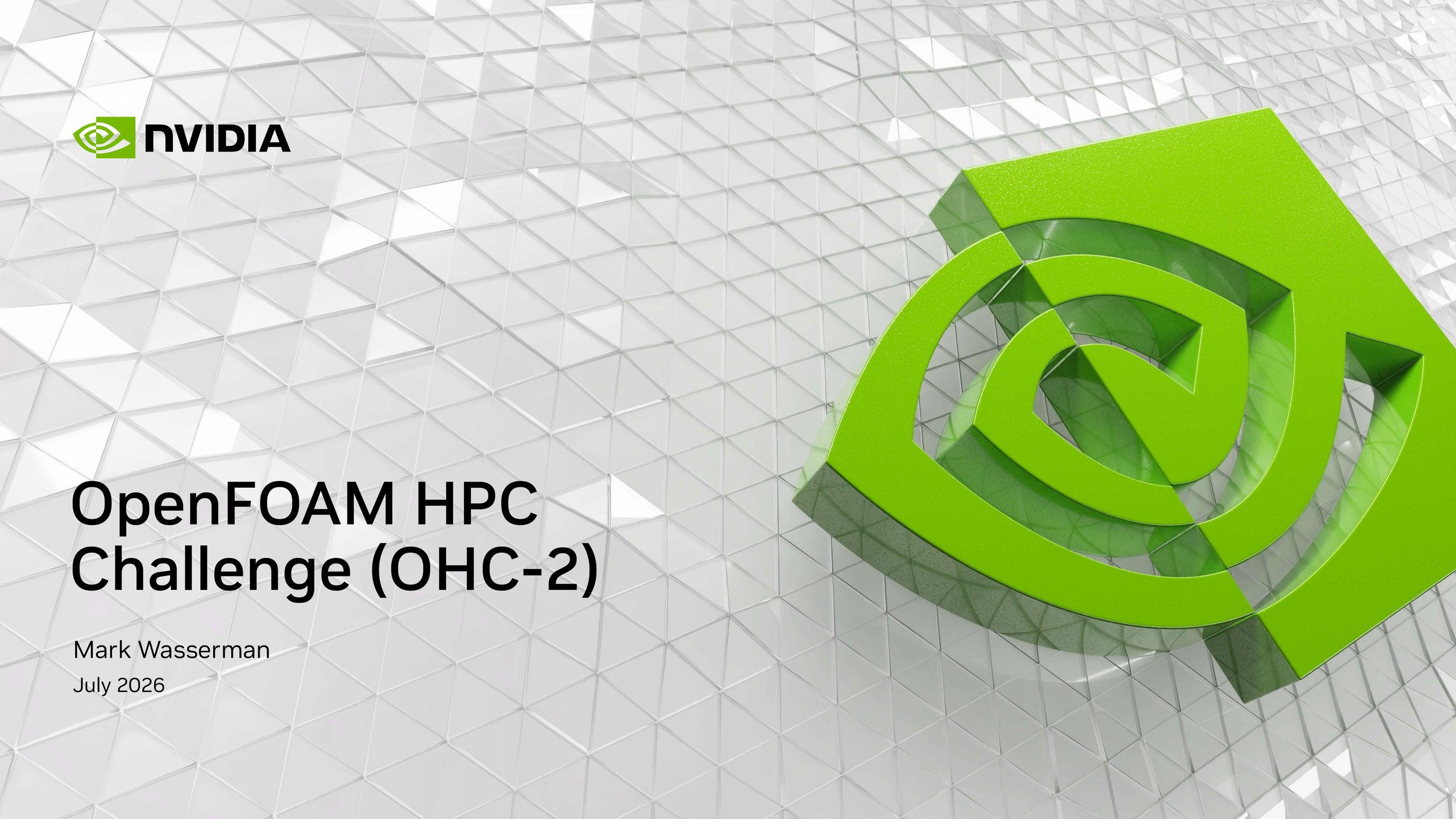


OpenFOAM HPC Challenge (OHC-2)

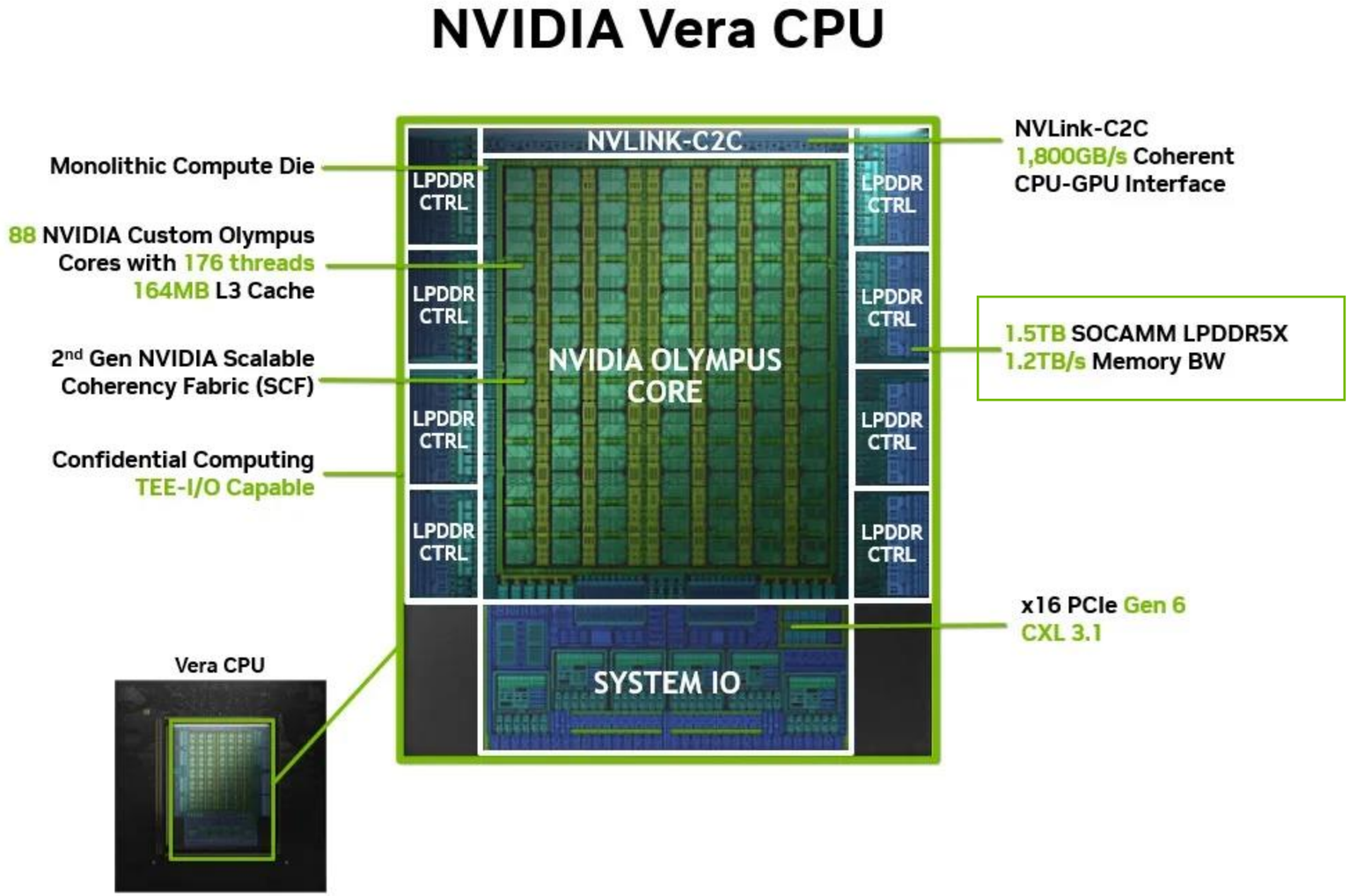
Mark Wasserman

July 2026

Introducing NVIDIA Vera: A Next-Generation CPU

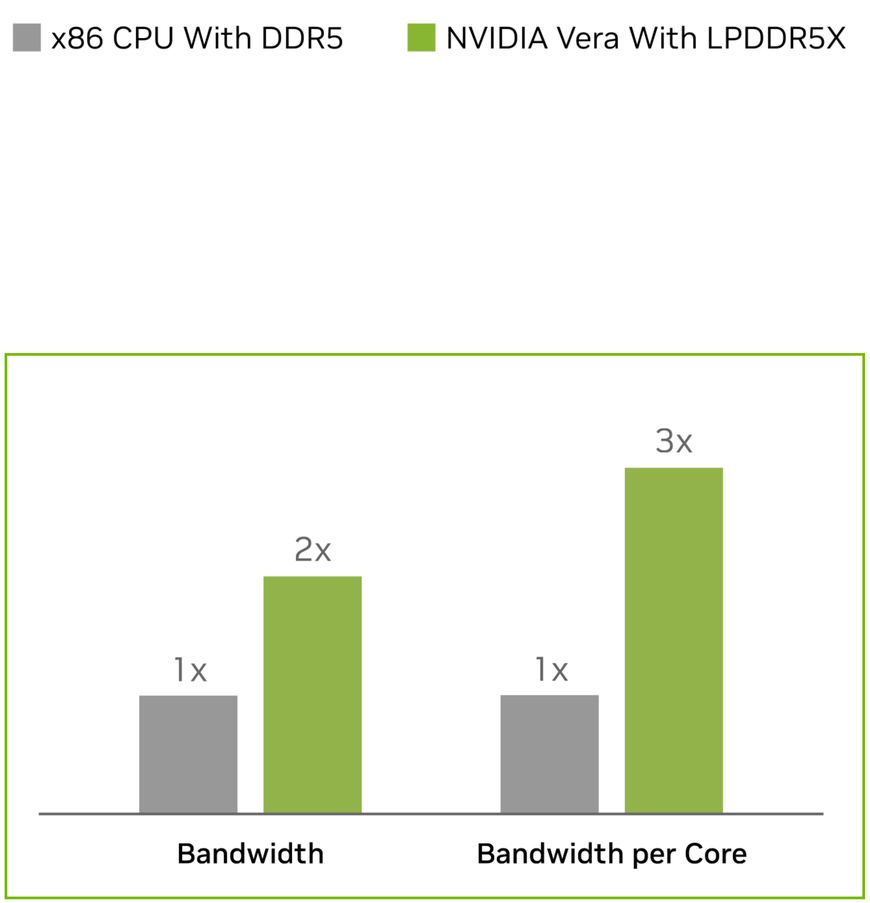
for AI Factories, Data Analytics and HPC Workloads

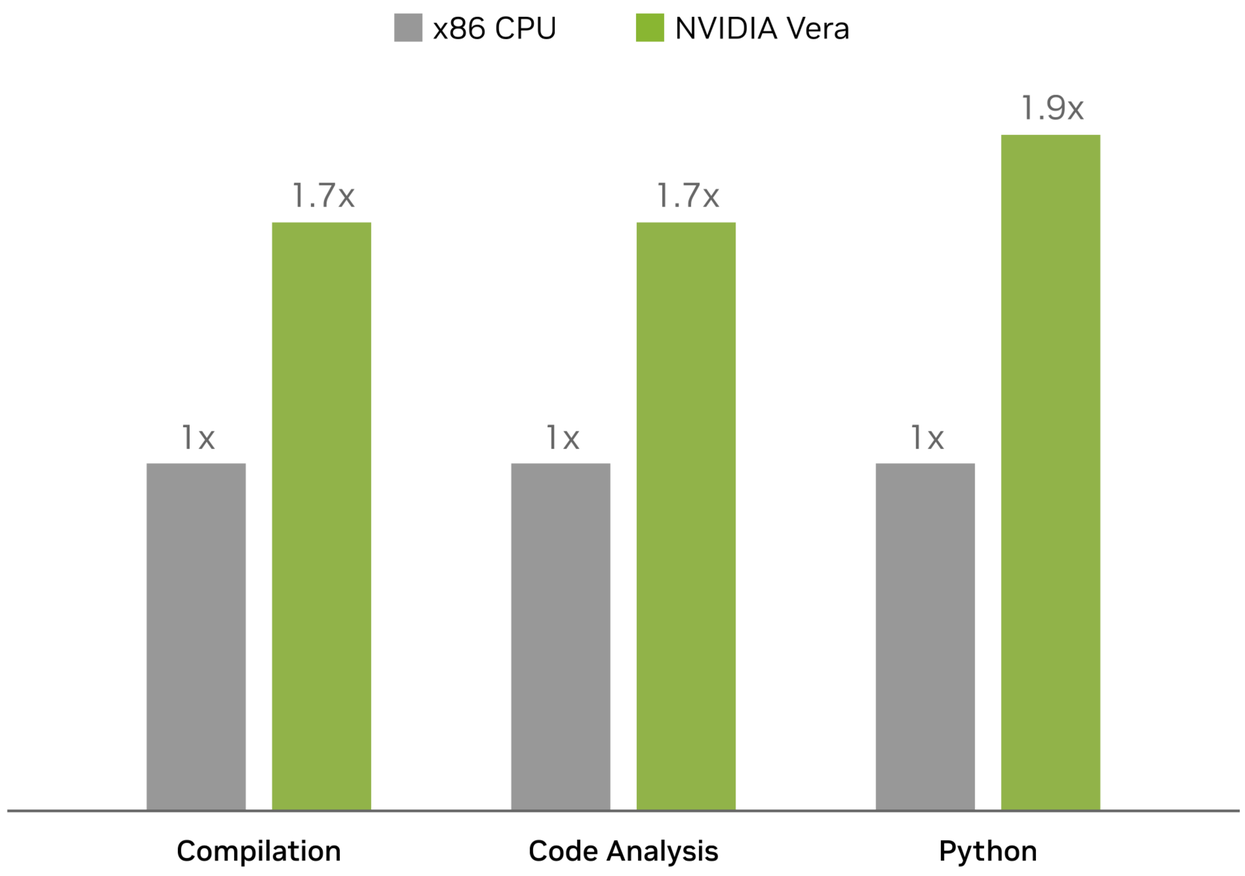




https://www.nvidia.com/en-us/data-center/vera-cpu/

Vera Performance Benchmarks

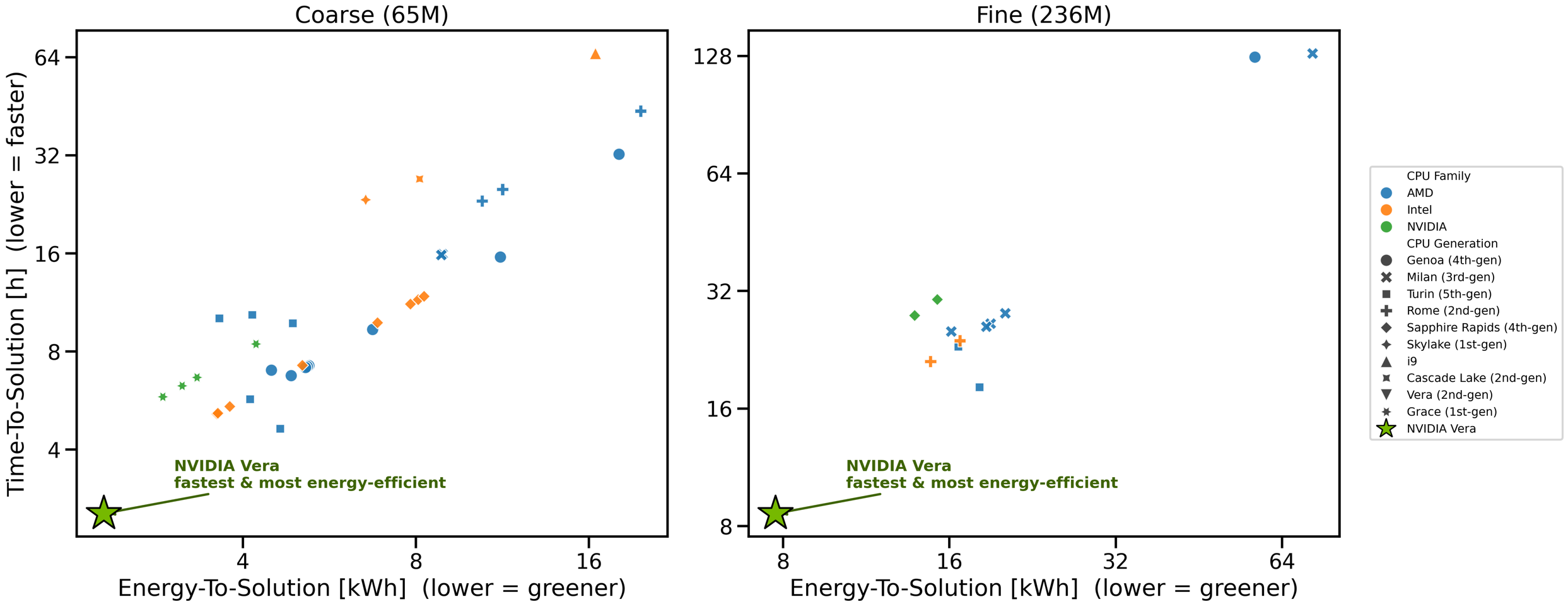




https://www.nvidia.com/en-us/data-center/vera-cpu/

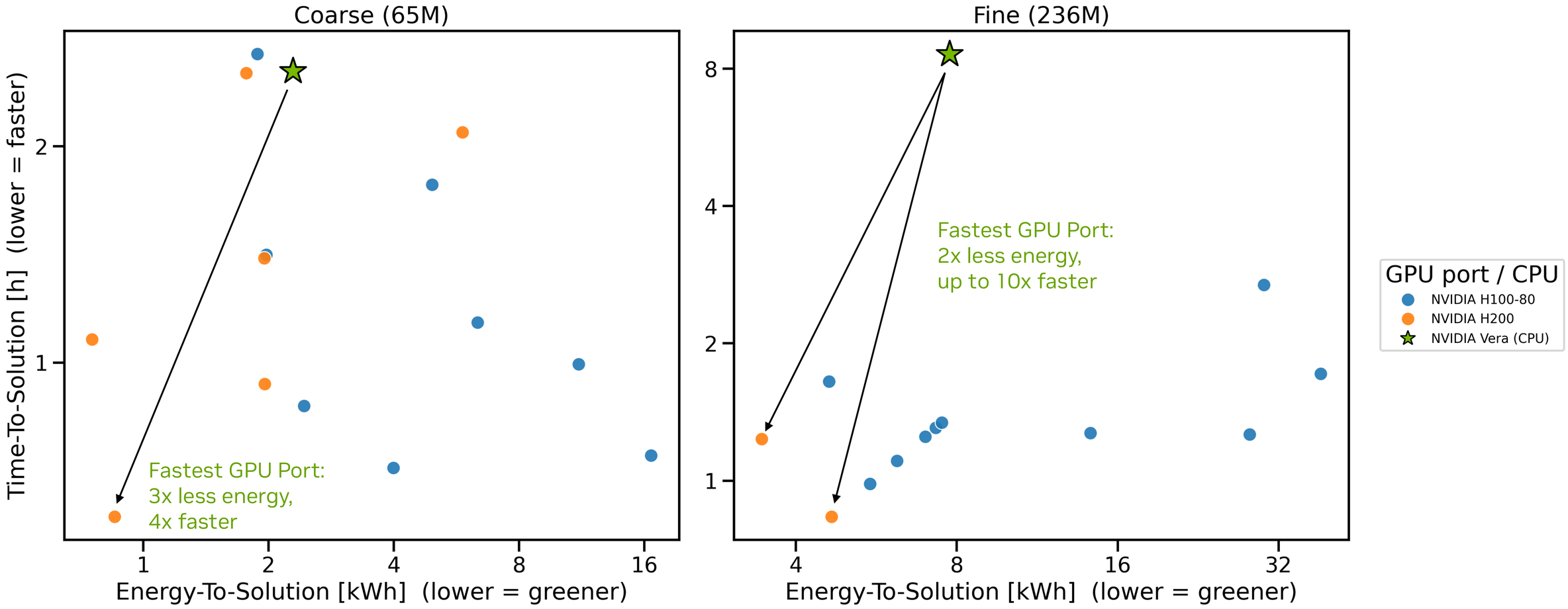
Vera OpenFOAM Performance – Single-Node

Vera provides rapid time-to-solution, with enhanced energy efficiency



Vera OpenFOAM Performance – Single-Node

OpenFOAM GPU ports already offer significant acceleration over CPU-based OF, with ample room for optimization



Conclusions

- **Vera is a next-generation custom ARM CPU** developed by NVIDIA, offering significant improvements in performance and power efficiency

- **Vanilla OpenFOAM (CPU-only) has best-in-class performance with Vera**
  - 2x speed-up and energy-efficiency improvements over other CPU results (x86, ARM)

- **OpenFOAM can be accelerated significantly if GPUs are leveraged**
  - OF GPU ports have made a huge leap over the past year
  - Up to 10x faster in 3x less energy w.r.t to CPU-only results (single-node)

NVIDIA